Project Proposal

Project Name: File Management System

Session: 2023-2027

(Reg.No.2023-UOk-03625, 2023-UOk-03626, 2023-UOk-03686)

Submitted to: Mr. Mehtab Mushtaq

Subject: OPERATIN SYSTEM

Submitted by:

Sadia (25), Zahra (26), Nayyab (65)

Date of Submission: January 30, 2026

DEPARTMENT OF CS & IT

FACULTY OF COMPUTER AND ENGINEERING

UNIVERSITY OF KOTLI AZAD JAMMU AND KASHMIR.

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1. Introduction

1.1 Overview

A File Management System is a core component of an operating system that manages how files are created, stored, accessed, modified, and deleted on a computer. It provides an organized way for users to handle data without needing to understand low-level hardware details. Through file management, users can safely store information, retrieve it when needed, and maintain data integrity.

This project focuses on developing a simple, console-based File Management System using C++. The system demonstrates fundamental file handling operations such as creating files, writing data, reading contents, and deleting files. By implementing this project, students gain practical experience with operating system concepts and understand how an OS interacts with the file system in a controlled and user-friendly manner.

1.2 Scope

The scope of this project is limited to basic file operations performed through a command-line interface. The system is designed for educational purposes and does not include advanced features such as file permissions, encryption, or graphical user interfaces. However, the project can be extended in the future to include these features.

2. Objectives of the Project

The main objectives of this project are:

- To understand the basic concepts of file management in operating systems.
- To implement file handling operations using C++.
- To provide a menu-driven interface for easy user interaction.
- To demonstrate how files are created, accessed, modified, and deleted.

- To improve programming skills related to system-level concepts.

3. Functional Requirements

3.1 Create File

The system shall allow the user to create a new file by entering a file name.

3.2 Write to File

The system shall allow the user to write data to a file and append new data to an existing file.

3.3 Read File

The system shall allow the user to read and display the contents of a specified file.

3.4 Delete File

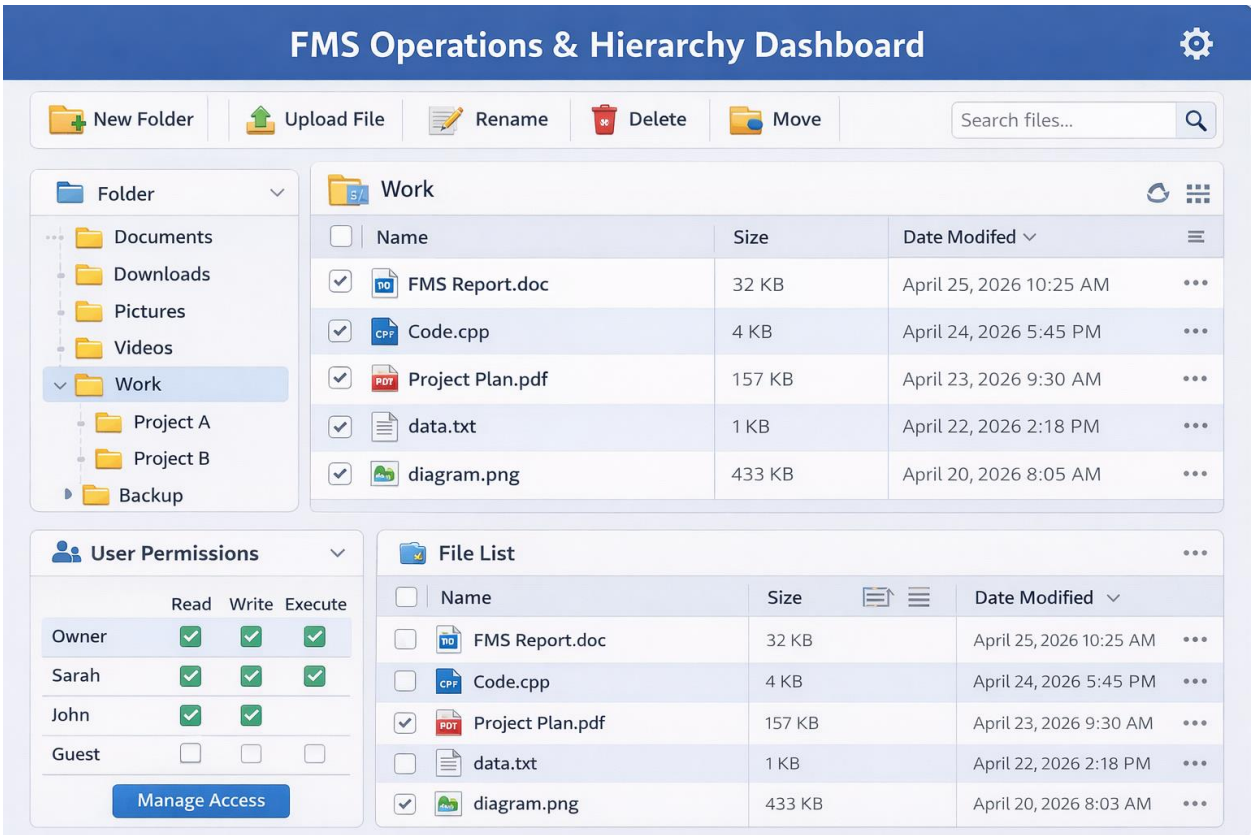
The system shall allow the user to delete an existing file from the system.

3.5 User Interaction

The system shall provide a menu-driven interface so users can easily select and perform file operations.

3.6 Error Handling

The system shall display appropriate error messages when a file does not exist or when an operation fails.



4. Non-Functional Requirements

4.1 Usability

The system shall be simple and easy to use, especially for beginner-level users.

4.2 Performance

The system shall perform file operations efficiently with minimal delay.

4.3 Reliability

The system shall handle invalid inputs gracefully without crashing.

4.4 Portability

The system shall be able to run on different operating systems that support C++.

4.5 Maintainability

The system shall be easy to modify and extend with additional features in the future.

4.6 Security

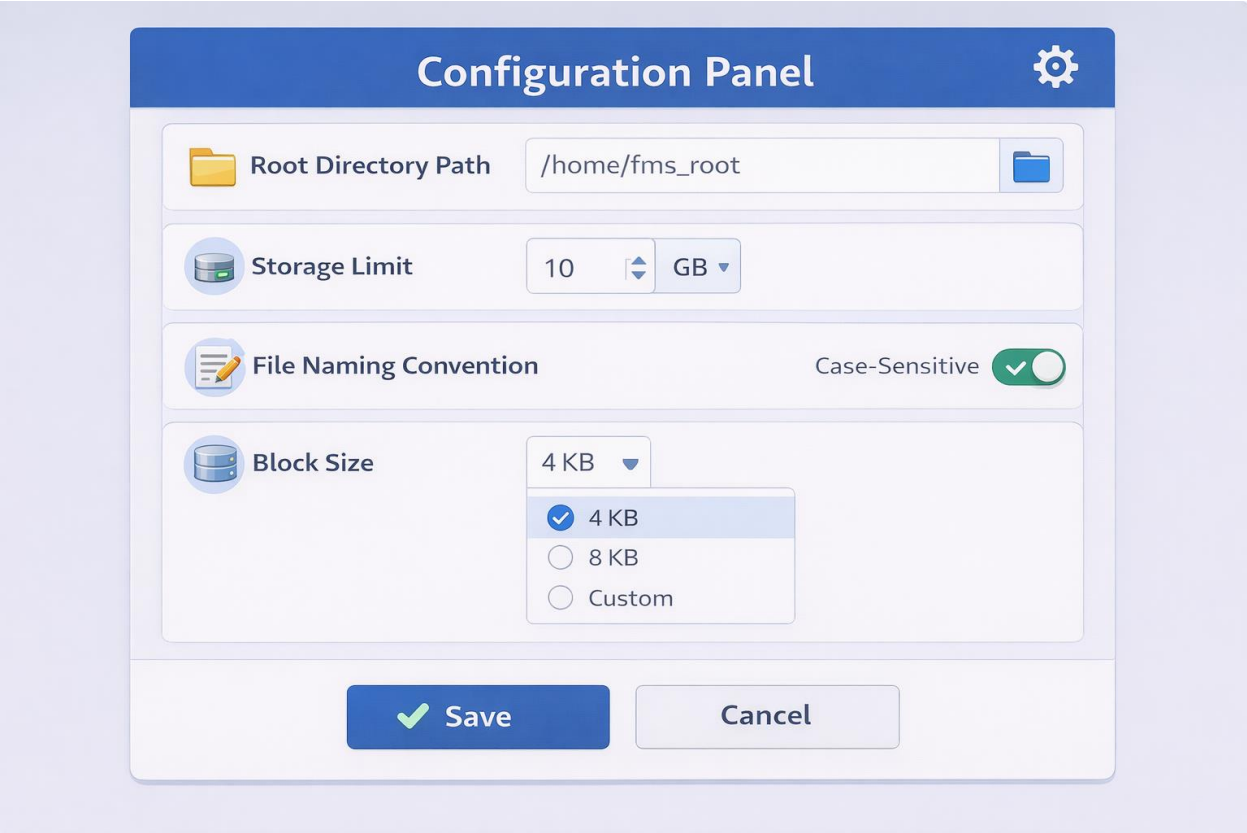
The system shall prevent invalid file operations through basic input validation.

5. Input Requirements

The File Management System requires certain inputs from the user in order to perform file operations. These inputs are provided through a menu-driven interface.

- File name provided by the user for creating, reading, writing, or deleting files.
- Text or data input that the user wants to store in a file.
- User selection of file operation such as create, read, write, or delete.
- Directory or path selection where the file operation will be performed.
- Configuration inputs such as storage limit, block size, and file naming rules for system setup.

These inputs allow the system to process user requests and manage files efficiently.



6. Output Requirements

The system produces outputs based on user actions and system operations.

- Display of file contents when a file is read successfully.
- Confirmation messages when a file is created, written, or deleted successfully.
- Error messages when a file does not exist or an invalid operation is attempted.
- Display of file list and directory structure for better understanding of stored files.
- Status messages related to system configuration and storage usage.

These outputs help users understand the results of their actions and ensure correct system behavior.

7. Assumptions & Constraints

Assumptions

- The user has basic knowledge of using a computer and command-line interface.
- The system will be used for educational and learning purposes only.
- The operating system supports C++ file handling libraries.
- Users will provide valid input most of the time.

Constraints

- The system is console-based and does not include a graphical user interface.
- Advanced features such as encryption and multi-user authentication are not implemented.
- The project operates within limited storage and system resources.
- Performance depends on the underlying operating system and hardware.

These assumptions and constraints define the boundaries within which the system operates.

6. References

1. Silberschatz, A., Galvin, P. B., & Gagne, G. Operating System Concepts. Wiley.

(File system structure, file operations, and directory management)

2. GeeksforGeeks. File System Concepts in Operating Systems.

(Reference material on file organization, access methods, and directory structures).

3. ChatGPT (Assistance in drafting, rewriting, and refining project content for clarity and structure)

4. Google Gemini (For illustrations and diagram generation)